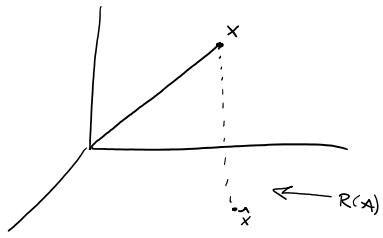


$A = [\, p_1 \cdots p_n \,]$

want: $A\underline{c} = \underline{x}$

choose $\hat{v}$ to minimize

$\| A\underline{c} - \underline{x} \|_2^2$

$A\underline{c} = \hat{\underline{x}}$

$\underline{c} = (A^H A)^{-1} A^H \underline{x}$

$A\underline{x} = \hat{\underline{x}} = \underbrace{A(A^H A)^{-1} A^H}_{P_A} \underline{x}$

$(A^H A)$

$$A = \begin{bmatrix} - & q_1^H & - \\ - & q_2^H & - \\ & \vdots & \\ - & q_n^H & - \end{bmatrix} \qquad A^H = \begin{bmatrix} | & | & & | \\ q_1 & q_2 & \cdots & q_n \\ | & | & & | \end{bmatrix}$$

$$A^H A = [\, q_1 \; q_2 \cdots \; q_n \,] \begin{bmatrix} q_1^H \\ q_2^H \\ \vdots \\ q_n^H \end{bmatrix} = \sum_{i=1}^{n} q_i q_i^H$$

$\left[\, I \,\right]^c \longrightarrow$

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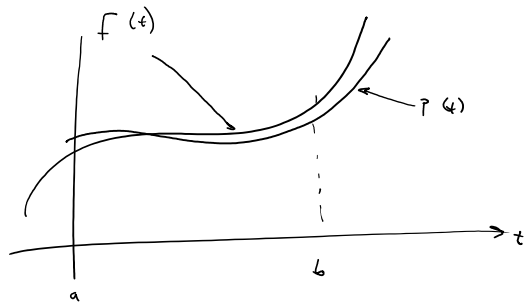



$f(t) = p(t) + e(t)$

want to choose $p(t)$ to make the error $e(t)$ "small."

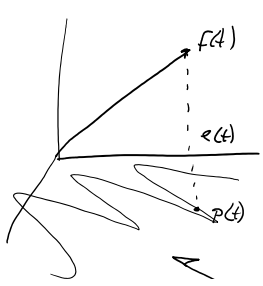
$p(t) = c_0 + c_1 t + c_2 t^2 + \cdots + c_{m-1} t^{m-1}$

$e(t) = f(t) - p(t)$

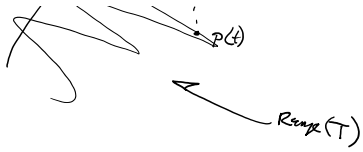
$\|e(t)\|_2^2 = \int_a^b e(t)^2 \, dt$

indeed

$\langle g(t), h(t) \rangle = \int_a^b g(t) h(t) \, dt$




$T = \{ 1, t, t^2, \ldots, t^{m-1} \}$

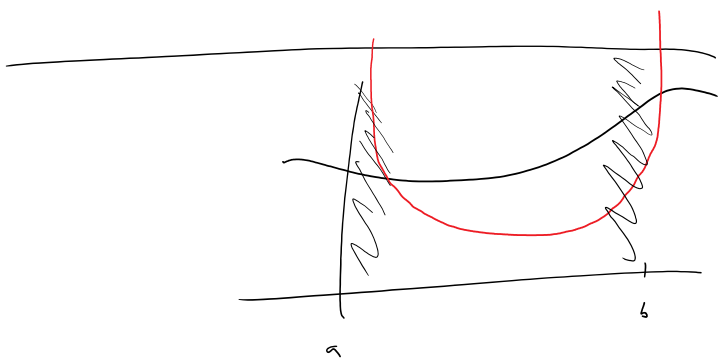



$f(t) - p(t) \perp$ "data"

$f(t) - p(t) \perp t^i \quad i = 0, 1, \ldots, m-1$

$\left[ \langle f(t) - p(t), t^i \rangle = 0 \quad i = 0, 1, \ldots, m-1 \right.$

$$\begin{bmatrix} \langle 1,1\rangle & \langle 1,t\rangle & \langle 1,t^2\rangle & \cdots & \langle 1,t^{m-1}\rangle \\ \langle t,1\rangle & \langle t,t\rangle & \langle t,t^2\rangle & \cdots & \langle t,t^{m-1}\rangle \\ \vdots & & & & \\ \langle t^{m-1},1\rangle & \langle t^{m-1},t\rangle & \cdots & & \langle t^{m-1},t^{m-1}\rangle \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_{m-1} \end{bmatrix} = \begin{bmatrix} \langle f,1\rangle \\ \langle f,t\rangle \\ \\ \\ \langle f,t^{m-1}\rangle \end{bmatrix}$$




$$\langle g(t), h(t) \rangle = \int_a^b w(t)\, g(t) h(t)\, dt$$

Now $\langle g(t), h(t) \rangle = \int_0^1 g(t) h(t)\, dt$

$$\langle t^i, t^j \rangle = \int_0^1 t^i t^j\, dt = \int_0^1 t^{i+j}\, dt = \frac{1}{i+j+1}$$

$$\begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} & \cdots & \frac{1}{m} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \cdots & \frac{1}{m+1} \\ \vdots & & & & \\ \frac{1}{m} & \frac{1}{m+1} & \cdots & & \frac{1}{2m} \end{bmatrix}$$

Hilbert matrix

$$\{1, t, t^2, \ldots, t^{m-1}\} \xrightarrow{GS} \{l_0, l_1(t), l_2(t), \ldots, l_{m-1}(t)\}$$

– orthogonal

$$\begin{bmatrix} \langle l_0, l_0\rangle & \langle l_0, l_1\rangle & \cdots & \langle l_0, l_{m-1}\rangle \\ & \ddots & & 0 \\ 0 & & \ddots & \\ & & & \langle l_{m-1}, l_{m-1}\rangle \end{bmatrix}$$

$$c_i = \frac{\langle f(t), l_i(t)\rangle}{\langle l_i(t), l_i(t)\rangle}$$

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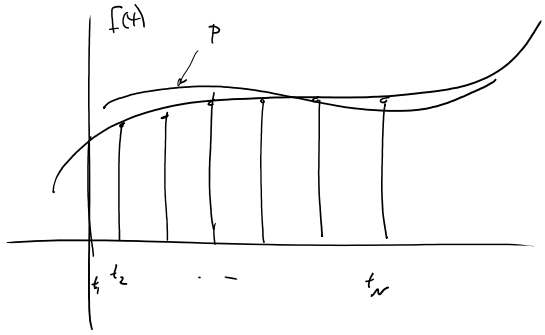


$f(t_i)$ $\quad i = 1, 2, \ldots, N$

$$p(t_i) = c_0 + c_1 t_i + c_2 t_i^2 + \cdots + c_{m-1} t_i^{m-1}$$

$$\min \sum_{i=1}^{N} \left(f(t_i) - p(t_i)\right)^2$$

$$\| f - p \|_2^2$$

$$\langle h, g\rangle = \sum_{i=1}^{n} h(t_i) g(t_i)$$

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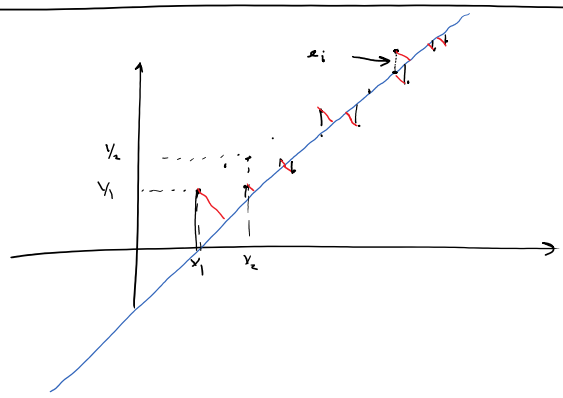
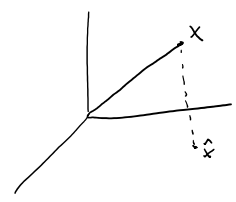



$(x_1, y_1)\ (x_2, y_2) \ldots (x_N, y_N)$

$$y_i = c_0 + c_1 x_i + e_i$$

$$\underbrace{\begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \\ 1 & x_N \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} c_0 \\ c_1 \end{bmatrix}}_{c} = \underbrace{\underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}}_{y} - \underbrace{\begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_N \end{bmatrix}}_{e}}_{\hat{y}}$$




$$\min \|e\|_2^2 = \sum_{i=1}^{N} e_i^2$$

$$\langle g, h\rangle = \sum_{i=1}^{N} g_i h_i$$

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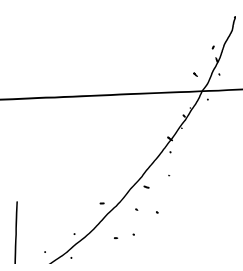
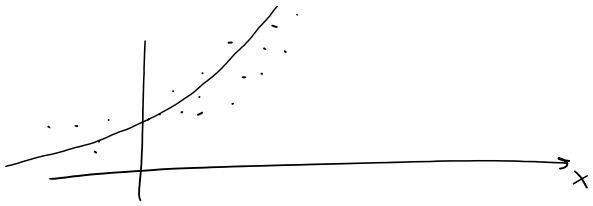


$$y_i \cong c_0 + c_1 x_i + c_2 x_i^2$$




$$y_i \cong c_0 \cdot 1 + c_1 x_i + c_2 x_i^2$$

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$$y_i \cong a e^{c x_i}$$

$$\ln y_i = \underbrace{c}_{c_1} \cdot x_i + \underbrace{\ln a}_{c_0 \cdot 1}$$

---

$$y_i = c a^{x_i}$$

$$\ln y_i = x_i \underbrace{\ln a}_{c_1} + \underbrace{\ln c}_{c_0}$$

---

$$y_i \cong c_0 + c_1 x_{1i} + c_2 x_{2i} + c_3 x_{1i} x_{2i} + c_4 x_{1i}^2 \ldots$$